

STOKKA, Norway

WMO: 011160

Lat: 65.961N

Lon: 12.471E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.2	-7.0	-17.9	0.8	-5.9	-15.6	1.0	-4.7	15.4	3.9	14.1	4.1	4.0	70	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	24.3	17.1	22.0	15.8	20.0	14.6	18.0	23.3	16.6	20.8	15.3	18.8	3.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.3	20.2	14.8	10.5	18.6	13.8	9.8	17.0	50.8	23.5	46.6	21.0	42.9	18.9	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.3	9.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-11.9	19.5	2.4	1.9	-13.6	20.9	-15.0	22.0	-16.4	23.1	-18.2	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	0.4	0.2	1.1	4.1	8.1	10.5	14.0	13.7	11.1	7.0	3.3	2.1		
	DBStd	5.95	3.83	3.62	3.19	3.15	3.67	3.03	3.56	2.90	2.97	3.16	3.54	3.31		
	HDD10.0	1713	299	275	277	179	83	26	2	2	21	102	202	246		
	HDD18.3	4394	557	508	535	426	318	236	143	148	216	352	451	504		
	CDD10.0	378	0	0	0	3	24	42	127	116	55	9	1	0		
	CDD18.3	17	0	0	0	0	1	1	11	4	0	0	0	0		
	CDH23.3	114	0	0	0	0	0	4	89	21	0	0	0	0		
	CDH26.7	17	0	0	0	0	0	0	16	1	0	0	0	0		
(14) Wind	WSAvg	4.4	5.0	4.9	5.1	4.4	4.1	4.0	3.5	3.6	3.8	4.5	4.8	5.3		
(15) Precipitation	PrecAvg	1380	131	120	115	91	75	83	90	94	156	154	133	153		
	PrecMax	1914	224	243	214	187	120	173	227	227	371	317	250	334		
	PrecMin	1050	0	19	33	24	21	15	30	15	40	35	11	0		
	PrecStd	264	61	64	53	42	24	43	58	47	84	81	77	91		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	7.8	9.5	18.2	23.2	24.0	28.7	25.9	21.1	17.0	12.9	9.7		
		MCWB	4.0	5.0	4.6	9.7	14.5	15.2	19.3	18.0	15.0	11.9	9.4	5.5		
	2%	DB	7.3	6.7	7.5	13.2	20.1	21.0	25.6	23.2	19.0	14.7	10.2	8.2		
		MCWB	5.5	4.8	4.7	7.5	12.9	14.2	18.5	16.6	13.4	10.2	7.5	6.2		
	5%	DB	6.3	5.9	6.5	10.9	17.5	18.2	22.9	21.0	17.7	12.9	9.2	7.5		
		MCWB	4.8	4.4	4.6	6.3	11.7	12.8	17.4	15.7	12.8	9.4	6.9	5.7		
	10%	DB	5.7	5.0	5.6	8.9	14.6	16.0	20.1	19.2	16.0	11.2	8.1	6.7		
		MCWB	4.2	3.4	3.8	5.3	10.1	11.9	15.5	14.6	12.2	8.8	6.1	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	6.4	6.7	9.9	15.3	15.9	20.6	18.9	16.1	12.2	10.0	8.1		
		MCDB	7.6	6.7	7.2	16.4	21.6	21.8	26.8	24.4	18.9	15.3	10.7	8.2		
	2%	WB	6.2	5.6	5.9	8.4	13.2	14.5	18.8	17.4	14.5	11.0	8.4	7.2		
		MCDB	6.6	6.1	6.6	12.4	18.9	19.7	24.4	22.0	17.7	13.2	9.8	7.6		
	5%	WB	5.2	4.7	5.0	6.9	11.9	13.4	17.5	16.1	13.6	10.1	7.4	6.2		
		MCDB	5.9	5.4	6.2	9.6	17.0	17.7	22.4	20.1	16.4	11.9	8.8	7.0		
	10%	WB	4.1	3.5	3.9	5.8	10.5	12.1	16.0	15.1	12.7	9.3	6.2	5.0		
		MCDB	5.1	4.5	5.5	8.1	14.5	15.4	19.5	18.4	15.2	10.9	7.8	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.0	4.4	5.2	6.0	5.4	5.9	5.9	5.6	4.5	4.0	4.1		
		MCDBR	4.3	4.3	4.7	8.4	9.8	9.5	10.9	9.8	8.5	6.2	4.8	4.8		
	5% WB	MCWBR	3.5	3.9	3.8	4.9	5.4	5.0	5.5	5.2	4.6	4.1	3.6	4.0		
		MCWBR	4.1	4.1	4.2	6.9	8.8	8.3	10.2	8.9	7.4	5.1	4.5	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.368	0.303	0.330	0.354	0.354	0.350	0.358	0.345	0.333	0.309	0.347	0.364		
		taud	2.526	2.186	2.273	2.343	2.432	2.488	2.490	2.517	2.520	2.337	2.489	2.477		
	Ebn at Noon		213	548	727	808	850	862	842	819	741	548	202	44		
		Edh at Noon	21	54	79	95	96	93	90	80	64	46	18	9		
(44) All-Sky Solar Radiation	RadAvg		0.07	0.44	1.68	3.32	4.49	4.89	4.49	3.46	1.85	0.72	0.14	0.01		
	RadStd		0.01	0.08	0.26	0.35	0.32	0.62	0.51	0.38	0.25	0.14	0.02	0.00		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (27 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	-93	-133	N/A	N/A	

Nomenclature: See separate page